

Symplectic Reduction for Cotangent Bundles

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1 Introduction

Symplectic reduction is a powerful method in differential geometry that allows for the simplification of a symplectic manifold while preserving its symplectic structure. It's particularly useful in the study of Hamiltonian systems, where it provides a systematic way to reduce the number of degrees of freedom by incorporating conserved quantities or symmetries. A common example arises in mechanics. The cotangent bundle T^*Q of a configuration space Q is a natural symplectic manifold. If a Lie group G acts on Q , it induces a Hamiltonian action on T^*Q with a moment map corresponding to conserved quantities. Symplectic reduction here simplifies the dynamics by restricting attention to the physical subspace allowed by the symmetries.

In this report, we will focus on the discussion of symplectic reduction of cotangent bundles. We first recall the connection form of a principle bundle, its associated curvature, and the momentum map on its cotangent bundle which is equivariant under the lifted action as background knowledge. In the second section, we state the general symplectic reduction theorem, then with acknowledging this theorem to be true, we prove the reduction of cotangent bundle at zero in details and generalise to the case for fixed points of coadjoint action.

2 Preliminary

Throughout this report, let Q be a principle bundle acted by a Lie group G on right, which is denoted by $Q \curvearrowright G$, and $\mathfrak{g}$ be the left Lie algebra of G , $r_g : Q \rightarrow Q$ the diffeomorphism of each group element g . Let $\text{AD}(g)$ be the left conjugation of $g \in G$, namely $\text{AD}(g)h = ghg^{-1}$ for every $h \in G$, which gives a left action of G on itself. Denote by $\text{Ad}(g)$ the tangent mapping at the identity and $\text{Ad}^*(g)$ its adjoint, which induces left and right actions on $\mathfrak{g}$ and $\mathfrak{g}^*$ respectively.

2.1 Connection of a Principle Bundle

We first review the connection form on a principle bundle and its associated curvature form as preliminary for our later discussion of cotangent bundle reduction.

Definition 2.1. A Ehresmann connection $\mathcal{A}$ of a principle bundle $Q \curvearrowright G$ is a smooth $\mathfrak{g}$ -valued 1-form satisfying the following two properties

1. $\mathcal{A}(\xi_Q) = \xi$, for $\forall \xi \in \mathfrak{g}$
2. $r_g^* \mathcal{A} = \text{Ad}(g^{-1}) \circ \mathcal{A}$, for $\forall g \in G$

Here, the $\xi_Q \in \mathfrak{X}(Q)$ is the infinitesimal generator of ξ , namely $\xi_Q(q) = \frac{d}{dt}|_{t=0} r_{\exp(t\xi)}(q)$, for every $q \in Q$.

2.1.1 The Vertical and Horizontal Distribution

Let $\pi : Q \rightarrow Q/G$ be the projection, call $\mathcal{V}_q := \ker \pi_q \subset T_q Q$ the vertical space at q and $\mathcal{H}_q := \ker \mathcal{A}(q)$ the horizontal space. One can immediately deduce from the first property that there is a direct sum decomposition for tangent space $T_q Q = \mathcal{V}_q \oplus \mathcal{H}_q$ since $\mathcal{V}_p \simeq \mathfrak{g}$ is related by the restriction of $\mathcal{A}(q)$ at the vertical space. And also we have

$\mathcal{H}_q \simeq T_{\pi(q)}Q/G$ related by the projection. The second property gives the right-invariant property of horizontal distribution. It is in fact a generalization of a corollary from the first property. To see that, let's compute

$$\begin{aligned} T_q r_g(\xi_Q(g)) &= \frac{d}{dt} \Big|_{t=0} r_{\exp(t\xi)g}(q) \\ &= \frac{d}{dt} \Big|_{t=0} r_{\text{Ad}(g^{-1}) \exp t\xi} \circ r_g(q) \\ &= \frac{d}{dt} \Big|_{t=0} r_{\exp(t\text{Ad}(g^{-1})\xi)} \circ r_g(q) \\ &= (\text{Ad}(g^{-1})\xi)_Q(r_g(q)) \end{aligned}$$

2.1.2 The Curvature of the Connection

By general theory of vector-valued differential forms (For more details, see [1]), we now introduce the curvature form of the connection, which will be of great importance in our later deduction.

Definition 2.2. The curvature form $\mathcal{B}$ of the connection $\mathcal{A}$ is a $\mathfrak{g}$ -valued 2-form defined by the following formula:

$$\mathcal{B} := d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]^\wedge$$

where the d is the exterior derivative, and $[\mathcal{A}, \mathcal{A}]^\wedge(w, v) := [\mathcal{A}(w), \mathcal{A}(v)] - [\mathcal{A}(v), \mathcal{A}(w)] = 2[\mathcal{A}(w), \mathcal{A}(v)]$ for $w, v \in TQ$ is the generalization of wedge product corresponding to the Lie bracket. It's not hard to see the the pull-back of $\mathcal{B}$ by right action has a similar formula to that of $\mathcal{A}$, namely

$$r_g^* \mathcal{B} = \text{Ad}(g^{-1}) \circ \mathcal{B}$$

Now consider the 1-form and 2-form induced by a dual vector $\mu \in \mathfrak{g}^*$ as follow

$$\begin{aligned} \alpha_\mu(w) &:= \langle \mu, \mathcal{A}(w) \rangle \\ \beta_\mu(w, v) &:= \langle \mu, \mathcal{B}(w, v) \rangle \end{aligned}$$

for $w, v \in TQ$. Further assume that μ is a fixed point under the coadjoint action, that is, μ 's stablizer $G_\mu := \{g \in G | \text{Ad}^*(g)\mu = \mu\}$ is the entire group G . Note that in this case α_μ is right-invariant:

$$\begin{aligned} r_g^* \alpha_\mu &= \langle \mu, r_g^* \mathcal{A} \rangle \\ &= \langle \mu, \text{Ad}(g^{-1}) \circ \mathcal{A} \rangle \\ &= \langle \text{Ad}^*(g^{-1})\mu, \mathcal{A} \rangle \\ &= \langle \mu, \mathcal{A} \rangle = \alpha_\mu \end{aligned}$$

The same argument goes for $\mathcal{B}$ and $\tilde{\beta}_\mu$. The right-invariant induces a unique 2-form B_μ on the quotient space Q/G by horizontal lift such that $\pi^* \tilde{\beta}_\mu = \beta_\mu$. Denote by $B_\mu = \pi_{Q/G}^* \tilde{\beta}_\mu$ the pull-back of B_μ to its cotangent bundle $T^*(Q/G)$, where $\pi_{Q/G} : T^*(Q/G) \rightarrow Q/G$ is the projection. Such relation can be summarized in the following diagram:

$$\begin{array}{c}
(Q, d\alpha_\mu = \beta_\mu) \\
\downarrow \pi \\
(Q/G, \tilde{\beta}_\mu) \xleftarrow{\pi_{Q/G}} (T^*(Q/G), B_\mu)
\end{array}$$

These terms will show up and play an important role when reducing at a non-zero fixed point of coadjoint action.

Now we prove a slightly not so trivial formula relating this two differential forms that will be useful later.

Proposition 2.3. *If $G_\mu = G$, then $d\alpha_\mu = \beta_\mu$.*

Proof. For $\forall \eta \in \mathfrak{g}$, $\langle \mu, \text{Ad}(g)\eta \rangle = \langle \text{Ad}^*(g)\mu, \eta \rangle = \langle \mu, \eta \rangle$ is a constant function on G , take derivative at $T_e G = \mathfrak{g}$ and use the fact that $T_e \text{Ad}(\xi) = \text{ad}_\xi = [\xi, \cdot]$ for any $\xi \in \mathfrak{g}$, we can deduce that $\langle \mu, [\xi, \eta] \rangle = 0$, which means μ sits in the annihilator of the derived algebra $[\mathfrak{g}, \mathfrak{g}]$. Then

$$\begin{aligned}
d\alpha_\mu &= \langle \mu, d\mathcal{A} \rangle \\
&= \langle \mu, \mathcal{B} \rangle - \frac{1}{2} \langle \mu, [\mathcal{A}, \mathcal{A}]^\wedge \rangle \\
&= \beta_\mu
\end{aligned}$$

since the term $\langle \mu, [\mathcal{A}, \mathcal{A}]^\wedge \rangle$ vanishes by the above argument. $\square$

2.2 Momentum Map on Cotangent Bundle

Let Q be a principle bundle with a symplectic form Ω acted by a Lie group G on the right symplectically. Recall that a momentum map $\mathbf{J} : Q \rightarrow \mathfrak{g}^*$ is a smooth map such that for every $\xi \in \mathfrak{g}$, $\xi_Q = X_{\langle \mathbf{J}, \xi \rangle}$, where ξ_Q is the infinitesimal generator defined as the last section and $X_{\langle \mathbf{J}, \xi \rangle}$ is the Hamiltonian vector field for the function $\langle \mathbf{J}, \xi \rangle \in \mathcal{F}(Q)$.

2.2.1 Regularity of Momentum Map on Symplectic Principle Bundle

We first show the regularity(i.e. the tangent mapping is surjective) of the momentum map. Before that, we need some preliminary on finite-dimensional symplectic linear algebra.

Definition 2.4. Let (V, Ω) be a finite-dimensional linear space with symplectic form Ω , $W \subset V$ a subspace of V , then the subspace

$$W^\Omega := \{v \in V \mid \Omega(v, w) = 0, \forall w \in W\}$$

is called the symplectic orthonormal space of W .

Using the non-degeneracy of Ω , we have the following conclusion:

Lemma 2.5. $\dim W + \dim W^\Omega = \dim V$, furthermore, $(W^\Omega)^\Omega = W$.

Proof. Let $W^\perp := \{f \in V^* | f|_W = 0\} \subset V^*$ be the annihilator of W , $r : V^* \rightarrow W^*$ be the restriction. It's obviously that $\ker r = W^\perp$, and since r is surjective by Zorn's lemma, we obtain that $\dim V = \dim V^* = \dim W^\perp + \dim W^* = \dim W^\perp + \dim W$. Next we show that Ω induces an isomorphism between W^Ω and $W^\perp$. In fact, define the flat operator

$$\flat : V \rightarrow V^* \\ v \mapsto v^\flat := \Omega(v, \cdot)$$

It's an isomorphism since Ω is non-degenerated. And clearly $(W^\Omega)^\flat \subset W^\perp$, so $\dim W^\Omega \leq \dim W^\perp$. Conversely, let the sharp operator $\sharp$ be the inverse mapping, by similar argument we obtain that $\dim W^\perp \leq \dim W^\Omega$, thus they share the same dimension, which complete the proof for the first assertion. Moreover, it's obvious that $W \subset (W^\Omega)^\Omega$, with the above conclusion we know that $\dim W = \dim (W^\Omega)^\Omega = \dim V - \dim W^\Omega$, hence $W = (W^\Omega)^\Omega$. $\square$

Proposition 2.6. *The momentum map $\mathbf{J}$ on a symplectic principle bundle $(Q, \Omega) \circlearrowright G$ is regular everywhere.*

Proof. For $\forall q \in Q$ and $\forall \dot{q} \in T_q Q$, pick a curve $q(t)$ such that $q(0) = q$, $\dot{q}(0) = \dot{q}$, for every $\xi \in \mathfrak{g}$

$$\begin{aligned} \langle T_q \mathbf{J} \dot{q}, \xi \rangle &= \langle \frac{d}{dt} \Big|_{t=0} \mathbf{J}(q(t)), \xi \rangle \\ &= \frac{d}{dt} \Big|_{t=0} \langle \mathbf{J}(q(t)), \xi \rangle \\ &= \Omega(\xi_Q, \dot{q}) \end{aligned}$$

Thus, $\ker T_q \mathbf{J} = (\mathcal{V}_q)^\Omega = \dim T_q Q - \dim \mathcal{V}_q = \dim Q - \dim \mathfrak{g}$, which completes the proof since $\dim \mathfrak{g}^* = \dim \mathfrak{g}$. $\square$

2.2.2 The Canonical Momentum Map on Cotangent Bundle

Let $Q \circlearrowright G$ be a principle bundle(not necessary to be symplectic), we shall show that there exist a nature momentum map on T^*Q corresponding to the lifted action which is symplectic with respect to the canonical 1-form Θ and the associated symplectic form $\Omega := -d\Theta$.

Let r_g be the right action of $g \in G$ on Q , define the lifted right action $T^*Q \circlearrowright G$ as $R_g := r_{g^{-1}}^*$. A quick observation is that the projection $\pi_Q : T^*Q \rightarrow Q$ is a covariant functor, that is $\pi_Q \circ R_g = r_g \circ \pi_Q$. The functoriality implies that the lifted action is indeed symplectic.

Proposition 2.7. *The lifted action is free and proper, thus makes the T^*Q to be a principle bundle. Moreover, the action is symplectic*

Proof. It's not hard to show that T^*Q is also a principle bundle. To see it's symplectic, take arbitrary $\alpha_q \in T^*Q$ and $g \in G$, one can obtain that

$$\begin{aligned} R_g^* \Theta(\alpha_q) &= \Theta(R_g \alpha_q) \circ T_{\alpha_q} R_g \\ &= \langle R_g \alpha_q, T_{R_g \alpha_q} \pi_Q \circ T_{\alpha_q} R_g \rangle \\ &= \langle R_g \alpha_q, T_{\pi_Q(\alpha_q)} r_g \circ T_{\alpha_q} \pi_Q \rangle \\ &= \langle \alpha_q, T_{\alpha_q} \pi_Q \rangle = \Theta(\alpha_q) \end{aligned}$$

$\square$

For every $\xi \in \mathfrak{g}$, denote its infinitesimal on Q and T^*Q by ξ_Q and ξ_{T^*Q} respectively. We can deduce the following two properties for the infinitesimal with above argument:

Property 2.8. ξ_{T^*Q} and ξ_Q are π_Q related and $\mathcal{L}_{\xi_{T^*Q}}\Theta = 0$.

Now let $\mathbf{J} : T^*Q \rightarrow \mathfrak{g}^*$ defined by $\langle \mathbf{J}(\alpha_q), \xi \rangle = \langle \alpha_q, \xi_Q(q) \rangle$ for each $\alpha_q \in q$. The above properties together with Cartan's magic formula implies that

$$\begin{aligned}\iota_{\xi_{T^*Q}}\Omega(\alpha_q) &= \iota_{\xi_{T^*Q}} \circ d\Theta(\alpha_q) \\ &= d \circ \iota_{\xi_{T^*Q}}\Theta(\alpha_q) - \mathcal{L}_{\xi_{T^*Q}}\Theta(\alpha_q) \\ &= d \langle \alpha_q, \xi_Q(q) \rangle = d \langle \mathbf{J}(\alpha_q), \xi \rangle\end{aligned}$$

Thus, $\xi_{T^*Q} = X_{\langle \mathbf{J}, \xi \rangle}$, which means $\mathbf{J}$ is indeed a momentum map. Moreover, using the conclusion in the last section, let's compute

$$\begin{aligned}\langle \mathbf{J}(R_g\alpha_q), \xi \rangle &= \langle R_g\alpha_q, \xi_Q(r_gq) \rangle \\ &= \langle r_{g^{-1}}^*\alpha_q, \xi_Q(r_gq) \rangle \\ &= \langle \alpha_q, T_{r_gq}r_{g^{-1}}(\xi_Q(r_gq)) \rangle \\ &= \langle \alpha_q, \text{Ad}(g)\xi_Q(q) \rangle \\ &= \langle \mathbf{J}(\alpha_q), \text{Ad}(g)\xi \rangle \\ &= \langle \text{Ad}^*(g)\mathbf{J}(\alpha_q), \xi \rangle\end{aligned}$$

This means $\mathbf{J} \circ R_g = \text{Ad}^*(g) \circ \mathbf{J}$, $\mathbf{J}$ is a functor from the single object category of T^*Q to $\mathfrak{g}$. Such relation is called equivariant.

3 Symplectic Reduction on Cotangent Bundle

With the preliminary in the last section, we shall now discuss the symplectic reduction on cotangent bundle. We first state the general symplectic reduction theorem on symplectic principle bundle, justifying from a perspective of symplectic algebra. Then we apply to the cotangent bundle of a principle bundle, which is a symplectic principle bundle with a canonical symplectic form and an equivariant momentum map. we first study the reduction at zero, and then generalise the result to those points whose stablizer is the entire group by a shifting trick.

3.1 General Symplectic Reduction Theorem

Let $(Q, \Omega) \curvearrowright G$ be a right principle bundle with symplectic form Ω and an equivariant momentum map $\mathbf{J}$. Denote by G_μ the stablizer of $\mu \in \mathfrak{g}^*$ under the coadjoint action. A first observation is that the action of G_μ on $\mathbf{J}^{-1}(\mu)$ is closed. Indeed,

$$\mathbf{J}(r_{g_\mu}(q_\mu)) = \text{Ad}^*(g_\mu)\mathbf{J}(q_\mu) = \text{Ad}^*(g_\mu)(\mu) = \mu$$

holds for arbitrary q_μ in the level set and g_μ in the associated stablizer. It's obvious the action is free and proper, thus it makes sense to define $Q_\mu := \mathbf{J}^{-1}(\mu)/G_\mu$ which is a principle bundle and consider the reduction on it.

Theorem 3.1 (Symplectic Reduction). Q_μ is a symplectic manifold with a unique symplectic form Ω_μ such that

$$\pi_\mu^* \Omega_\mu = i_\mu^* \Omega$$

Here, $\pi_\mu : \mathbf{J}^{-1}(\mu) \rightarrow Q_\mu$ is the projection and $i_\mu : \mathbf{J}^{-1}(\mu) \rightarrow Q$ the inclusion.

Note that the uniqueness obviously holds once the existence is verified since the projection π_μ is a surjective submersion. Let's make a draft of how this conjecture derives. First note that $\mathbf{J}^{-1}(\mu)$ is an embedded submanifold since $\mathbf{J}$ is regular everywhere. $i_\mu^* \Omega$ is the restriction of Ω to the tangent space of $\mathbf{J}^{-1}(\mu)$, which, by a corollary in the proof of 2.6, is nothing but the symplectic orthonormal space of the vertical space. So if $\mathcal{V}_{q_\mu} \cap \mathcal{V}_{q_\mu}^\Omega \neq \{0\}$ for some $q_\mu \in \mathbf{J}^{-1}(\mu)$, then the pull-back of Ω will degenerate. To remove the degeneracy we need to find out and kill the intersection by quotient. To see this, it's helpful to consider the linear case.

Lemma 3.2. Let V be a finite dimensional vector space with a anti-symmetry form Ω , and $\flat : V \rightarrow V^*, v \mapsto \Omega(v, \cdot)$ the flat operator. Then there exists a unique symplectic form $\tilde{\Omega}$ on the quotient space $V/\ker \flat$ such that $\pi^* \tilde{\Omega} = \Omega$, here π is the projection.

Proof. The uniqueness is obvious since π is surjective. To see existence, let $[v]$ be the equivariant class of $v \in V$ in the quotient space. Define $\tilde{\Omega}(v, w) := \Omega([v], [w])$. Such a $\tilde{\Omega}$ is well-defined since for any $[v'] = v$, $[w'] = [w]$, $v - v'$ and $w - w'$ sit in the kernel, therefore

$$\Omega(v, w) - \Omega(v', w') = \Omega(v - v', w) - \Omega(w - w', v') = 0 - 0 = 0$$

Ω is the pull-back of $\tilde{\Omega}$ by construction. It remains to show non-degeneracy. Pick any $[v] \in \ker \flat$ and arbitrary $[w]$, $0 = \tilde{\Omega}([v], [w]) = \Omega(v, w)$ implies $v \in \ker \flat$ since w runs over V as $[w]$ runs over $V/\ker \flat$. $\square$

The task now is to find out the kernel of the flat operator for the restriction of Ω on $\mathbf{J}^{-1}(\mu)$. We claim that it's the span of the infinitesimal of G_μ 's Lie algebra $\mathfrak{g}_\mu$.

Proposition 3.3. $\mathcal{V}_q \cap \mathcal{V}_q^\Omega = \{\xi_Q(q) | \xi \in \mathfrak{g}_{\mathbf{J}(q)}\}$

Proof. For arbitrary ξ and $\eta \in \mathfrak{g}$, by definition of momentum map, we have

$$\begin{aligned} \Omega(\xi_Q(q), \eta_Q(q)) &= d \langle \mathbf{J}, \xi \rangle (\eta_Q(q)) \\ &= \frac{d}{dt} \Big|_{t=0} \langle \mathbf{J}(r_{\exp(t\eta)}(q)), \xi \rangle \end{aligned}$$

If $\eta \in \mathfrak{g}_{\mathbf{J}(q)}$, then $\mathbf{J}(r_{\exp(t\eta)}(q)) = \mathbf{J}(q)$ by a corollary of equivariance, thus $\Omega(\xi_Q(q), \eta_Q(q)) = 0$ for every ξ , $\eta \in \mathcal{V}_q \cap \mathcal{V}_q^\Omega$; On the other hand, If $\eta \in \mathcal{V}_q \cap \mathcal{V}_q^\Omega$, then $\mathbf{J}(r_{\exp(t\eta)}(q))$ must be a constant as ξ runs over $\mathfrak{g}$, take $t = 0$, one can obtain the constant is nothing but $\mathbf{J}(q)$, thus $\eta \in \mathfrak{g}_{\mathbf{J}(q)}$, this completes the proof. $\square$

Now we know that $Q_\mu = \mathbf{J}^{-1}(\mu)/G_\mu$ actually kills the degeneracy by the above argument and the induced non-degenerated 2-form is close since exterior differential operator commutes with pull-back, thus is a symplectic form satisfying the formula characterising it. In fact, we have the following bundle morphism:

$$\begin{array}{ccc}
\mathcal{V}^\Omega & \xrightarrow{Tr_{g_\mu}} & \mathcal{V}^\Omega \\
\downarrow T\pi_\mu & & \downarrow T\pi_\mu \\
TQ_\mu & \xrightarrow{f_{g_\mu}} & TQ_\mu
\end{array}$$

Here, $f_{g_\mu} : TQ_\mu \rightarrow TQ_\mu$ is the uniquely induced bundle isomorphism for given $g_\mu \in G_\mu$, one can check it's even symplectic with respect to the two symplectic form induced by the two tangent mapping $T\pi_\mu$ since the upstairs is symplectic linear transformation. Thus there does exist a well-defined symplectic form on the quotient space.

3.2 Reduction on Cotangent Bundle

Throughout the section, let $Q \circlearrowleft G$ be a principle bundle with connection $\mathcal{A}$ and curvature $\mathcal{B}$, α_μ , β_μ and B_μ defined as in 2.1.2 and Q_μ as in the last section for $\mu \in \mathfrak{g}^*$, denote the projections by $\pi : Q \rightarrow Q/G$, $\pi_\mu : \mathbf{J}^{-1} \rightarrow \mathbf{J}^{-1}(\mu)/G_\mu =: (T^*Q)_\mu$, $\pi_Q : T^*Q \rightarrow Q$ and $\pi_{Q/G} : T^*(Q/G) \rightarrow Q/G$ respectively. Here $\mathbf{J}$ is the canonical momentum map of T^*Q with its canonical 1-form Θ and symplectic form Ω , the canonical 1-form and symplectic form on $T^*(Q/G)$ denoted by θ and ω respectively.

3.2.1 Reduction at Zero

For the case of $\mu = 0$, we first construct a symplectic diffeomorphism $\varphi_0 : ((T^*Q)_0, \Omega_\mu) \rightarrow (T^*(Q/G), \omega)$. The strategy is to first find a G -invariant mapping on $\mathbf{J}^{-1}(0)$ and induce a map on the quotient space. Let $\bar{\varphi}_0 : \mathbf{J}^{-1}(0) \rightarrow T^*(Q/G)$ satisfy that for arbitrary $\alpha_{q_0} \in \mathbf{J}^{-1}(0)$ with $\pi_Q(\alpha_{q_0}) = q_0$ and $v_{q_0} \in T_{q_0}Q$, $\langle \bar{\varphi}_0(\alpha_{q_0}), T_{q_0}\pi(v_{q_0}) \rangle = \langle \alpha_{q_0}, v_{q_0} \rangle$. It's well-defined since if $T_{q_0}\pi(v_{q_0}^1) = T_{q_0}\pi(v_{q_0}^2)$ for some $v_{q_0}^1, v_{q_0}^2 \in T_{q_0}Q$, there exists a unique $\xi \in \mathfrak{g}$ such that $v_{q_0}^1 - v_{q_0}^2 = \xi_Q(q_0)$, thus $\langle \alpha_{q_0}, v_{q_0}^1 - v_{q_0}^2 \rangle = \langle \alpha_{q_0}, \xi_Q(q_0) \rangle = \langle \mathbf{J}(\alpha_{q_0}), \xi \rangle = \langle 0, \xi \rangle = 0$. That $\bar{\varphi}_0$ is G -invariant can be seen through the following computation using $\pi = \pi \circ r_g$:

$$\begin{aligned}
\langle \bar{\varphi}_0(R_g \alpha_{q_0}), T_{r_g q_0} \pi(v_{r_g q_0}) \rangle &= \langle R_g \alpha_{q_0}, v_{r_g q_0} \rangle \\
&= \langle \alpha_{q_0}, T_{r_g q_0} r_{g^{-1}}(v_{r_g q_0}) \rangle \\
&= \langle \bar{\varphi}_0(\alpha_{q_0}), T_{q_0} \pi \circ T_{r_g q_0} r_{g^{-1}}(v_{r_g q_0}) \rangle \\
&= \langle \bar{\varphi}_0(\alpha_{q_0}), T_{r_g q_0} (\pi \circ r_{g^{-1}})(v_{r_g q_0}) \rangle \\
&= \langle \bar{\varphi}_0(\alpha_{q_0}), T_{r_g q_0} \pi(v_{r_g q_0}) \rangle
\end{aligned}$$

Therefore, there is a unique induced map φ_0 on the reduction space that is characterised by the formula: $\bar{\varphi}_0 = \varphi_0 \circ \pi$.

Now we show φ_0 is a symplectic diffeomorphism. To show it's surjective it suffices to show that $\bar{\varphi}_0$ is surjective. In fact, $\mathbf{J}^{-1}(0) = \mathcal{V}^\perp$, that is, its the subbundle of the cotangent bundle consisting of all the annihilator of all vertical space. For arbitrary equivariant class $[q]$ of $q \in Q$ and $\alpha_{[q]} \in T_{[q]}^*(Q/G)$, we can find an $\alpha_q \in T_q^*Q$ defined by $\langle \alpha_q, v_q \rangle := \langle \alpha_{[q]}, T_q \pi(v_q) \rangle$ for $\forall v_q \in T_q Q$, one can immediately verify that $\bar{\varphi}_0(\alpha_q) = \alpha_{[q]}$. To see it's injective, notice that $\bar{\varphi}_0 : \mathbf{J}^{-1}(0) = \mathcal{V}^\perp \rightarrow T^*(Q/G)$ is actually a bundle epimorphism by the surjectivity argument, while these two bundle share the

same rank of $\dim Q - \dim G$, this implies $\bar{\varphi}_0$ is isomorphism when restricted to fiber, since π_0 kills the symmetry, φ_0 must be injective.

To see φ_0 is symplectic, since π_0 is submersion and hence π_0^* is injective, it suffices to show $\pi_0^*\varphi_0^*\omega = \pi_0^*\Omega$. By general reduction theorem, that is to show $\bar{\varphi}_0^*\omega = i_0^*\Omega$. Notice that $\pi \circ \pi_Q = \pi_{Q/G} \circ \bar{\varphi}_0$, pick arbitrary $\alpha_q \in \mathbf{J}^{-1}(0)$ with base point $q \in Q$, $\dot{\alpha}_q$ a vector on α_q , one can compute that

$$\begin{aligned}\bar{\varphi}_0^*\omega(\dot{\alpha}_q) &= \omega(T_{\alpha_q}\bar{\varphi}_0\dot{\alpha}_q) \\ &= \langle \bar{\varphi}_0(\alpha_q), T_{\bar{\varphi}_0(\alpha_q)}\pi_{Q/G} \circ T_{\alpha_q}\bar{\varphi}_0\dot{\alpha}_q \rangle \\ &= \langle \bar{\varphi}_0(\alpha_q), T_{\alpha_q}(\pi_{Q/G}, \bar{\varphi}_0)\dot{\alpha}_q \rangle \\ &= \langle \bar{\varphi}_0(\alpha_q), T_{\alpha_q}(\pi \circ \pi_Q)\dot{\alpha}_q \rangle \\ &= \langle \alpha_q, T_q\pi_Q\dot{\alpha}_q \rangle \\ &= i_0^*\Omega(\dot{\alpha}_q)\end{aligned}$$

Since φ_0 is symplectic by the above computation, it must be a immersion, while its bijective, it's in fact a symplectic diffeomorphism.

3.2.2 Reduction at Fixed Point of Codajoint Action

Now assume that $\mu \in \mathfrak{g}^*$ has the stablizer $G_\mu = G$, there is an induced shifting mapping relating its level set with that of zero, hence relating their quotient space. First observer that

$$\begin{aligned}\langle \mathbf{J}(\alpha_\mu(q_\mu)), \xi \rangle &= \langle \alpha_\mu(q_\mu), \xi_Q(q_\mu) \rangle \\ &= \langle \mu, \mathcal{A}(\xi_Q(q)) \rangle \\ &= \langle \mu, \xi \rangle\end{aligned}$$

That is, $\mathbf{J}(\alpha_\mu) = \mu$. Therefore we can introduce the diffeomorphism:

$$\begin{aligned}\text{shift}_\mu : \mathbf{J}^{-1}(\mu) &\rightarrow \mathbf{J}^{-1}(0) \\ \gamma_{q_\mu} &\mapsto \gamma_{q_\mu} - \alpha_\mu(q_\mu)\end{aligned}$$

which is the restriction on the level set of the shifting mapping on cotangent bundle:

$$\begin{aligned}\text{Shift}_\mu : T^*Q &\rightarrow T^*Q \\ \gamma(q) &\mapsto \gamma(q) - \alpha_\mu(q)\end{aligned}$$

By computation in 2.1.2 we know that α_μ is right invariant, thus one can obtain that

$$\text{shift}_\mu(R_g(\gamma_{q_\mu})) = R_g(\gamma_{q_\mu}) + \alpha_\mu(r_g(q_\mu)) = R_g(\gamma_{q_\mu} + \alpha_\mu(q_\mu)) = R_g(\text{shift}_\mu(\gamma_{q_\mu}))$$

This means the shifting map preserves fibers, and hence induces a diffeomorphism $\widetilde{\text{shift}}_\mu$ between the quotient spaces such that the following diagram commutes:

$$\begin{array}{ccc}
T^*Q & \xrightarrow{\text{Shift}_\mu} & T^*Q \\
i_\mu \uparrow & & i_0 \uparrow \\
\mathbf{J}^{-1}(\mu) & \xrightarrow{\text{shift}_\mu} & \mathbf{J}^{-1}(0) \\
\downarrow \pi_\mu & & \downarrow \pi_0 \\
Q_\mu & \xrightarrow{\widetilde{\text{shift}_\mu}} & Q_0
\end{array}$$

Eventhough $\widetilde{\text{shift}_\mu}$ is a nice diffeomorphism which induces a diffeomorphism $\varphi_\mu := \varphi_0 \circ \widetilde{\text{shift}_\mu} : Q_\mu \rightarrow T^*(Q/G)$, notice that $\widetilde{\text{shift}_\mu}$ may not be symplectic, thus there should be an extra term $\omega' \in \Omega^2(T^*(Q/G))$ such that $\varphi_\mu : ((T^*Q)_\mu, \Omega_\mu) \rightarrow (T^*(Q/G), \omega + \omega')$ is symplectic diffeomorphism, where Ω_μ is the symplectic form by theorem of Symplectic Reduction.

To compute the extra term, first compute the pull-back of Shift. Pick any $\dot{\alpha}_q \in T_{\alpha_q} T^*Q$ for $\alpha_q \in T_q^*Q$, using the fact that $\pi_Q \circ \alpha_\mu \circ \pi_Q = \pi_Q$, then one can obtain

$$\begin{aligned}
\text{Shift}^* \Theta(\dot{\alpha}_q) &= \Theta(\dot{\alpha}_q) - \Theta(T_{\alpha_\mu(q)} \alpha_\mu \circ T_{\alpha_q} \pi_Q \dot{\alpha}_q) \\
&= \Theta(\dot{\alpha}_q) - \langle \alpha_\mu(q), T_{\alpha_q} (\pi_Q \circ \alpha_\mu \circ \pi_Q) \dot{\alpha}_q \rangle \\
&= \Theta(\dot{\alpha}_q) - \langle \alpha_\mu(q), T_{\alpha_q} \pi_Q \dot{\alpha}_q \rangle \\
&= (\Theta - \pi_Q^* \alpha_\mu)(\dot{\alpha}_q)
\end{aligned}$$

As a corollary, we have

$$\begin{aligned}
\text{Shift}^* \Omega &= -\text{Shift}^* d\Theta \\
&= -d\text{Shift}^*(\Theta) \\
&= -d\Theta + d\pi_Q^* \alpha_\mu \\
&= \Omega + \pi_Q^* d\alpha_\mu \\
&= \Omega + \pi^* \beta_\mu
\end{aligned}$$

Now we can try to discover the explict expression for ω' . By reduction theorem we have

$$\begin{aligned}
i_\mu^* \Omega &= \pi_\mu^* \Omega_\mu \\
&= \pi_\mu^* \circ \widetilde{\text{shift}_\mu}^* \circ \varphi_0^* (\omega + \omega') \\
&= \text{shift}_\mu^* \circ \pi_0^* (\Omega_0 + \varphi_0^* \omega') \\
&= \text{shift}_\mu^* (i_0^* \Omega + \pi_0^* \varphi_0^* \omega') \\
&= i_\mu^* \circ \text{Shift}_\mu^* \Omega + \text{shift}_\mu^* \pi_0^* \varphi_0^* \omega' \\
&= i_\mu^* \Omega + i_\mu^* \pi_Q^* \beta_\mu + \text{shift}_\mu^* \pi_0^* \varphi_0^* \omega'
\end{aligned}$$

Notice that $\pi_Q \circ i_0 \circ \text{shift}_\mu = \pi_Q \circ i_\mu$, one can obtain

$$i_0^* \circ \pi_Q^* \beta_\mu + \pi_0^* \circ \varphi_0^* \omega' = 0$$

since shift_μ cancels because it's invertible. By the construction of φ_0 , one may check the following diagram commutes:

$$\begin{array}{ccccc} T^*Q & \xrightarrow{\pi_Q} & Q & \xrightarrow{\pi} & Q/G \\ i_0 \uparrow & & & & \uparrow \pi_{Q/G} \\ \mathbf{J}^{-1}(0) & \xrightarrow{\pi_0} & \mathbf{J}^{-1}(0)/G & \xrightarrow{\varphi_0} & T^*(Q/G) \end{array}$$

And by the relation of β_μ and B_μ that had been explored, one can obtain

$$\begin{aligned} 0 &= i_0^* \circ \pi_Q^* \circ \pi^* \tilde{\beta}_\mu + \pi_0^* \circ \varphi_0^* \omega' \\ &= \pi_0^* \circ \varphi_0^* \circ \pi_{Q/G}^* \tilde{\beta}_\mu + \pi_0^* \circ \varphi_0^* \omega' \\ &= \pi_0^* \circ \varphi_0^* (B_\mu + \omega') \end{aligned}$$

Since π_0 is a submersion, π_0^* is injective pointwise, and φ_0 is diffeomorphism, we finally obtain

$$\omega' = -B_\mu$$

3.2.3 Reduction at Arbitrary Point

In the last section we state the idea of how to do reduction with $\mu \in \mathfrak{g}$ whose stablizer G_μ is a proper subset of G . First notice that G_μ is a close subgroup, by Cartan's closed subgroup theorem, G_μ is a Lie group. Denote the associated momentum map by $\mathbf{J}_\mu$ and the restriction of μ to $\mathfrak{g}_\mu$ by μ' . One can construct a new principle bundle $Q \circlearrowleft G_\mu$ with connection form $\mathcal{A}^\mu$ (which could be induced by some other structure such as a Riemannian metric) and curvature $\mathcal{B}^\mu$. By the last section we have

$$(\mathbf{J}_\mu^{-1}(\mu')/G_\mu, \Omega_{\mu'}) \cong (T^*(Q/G_\mu), \Omega - B_{\mu'})$$

while there is a nature inclusion $\mathbf{J}^{-1}(\mu) \hookrightarrow \mathbf{J}_\mu^{-1}(\mu)$, which can be induced to the quotient space modulo by G_μ . This gives us the embedding case:

$$(\mathbf{J}^{-1}(\mu), \Omega_\mu) \hookrightarrow (T^*(Q/G_\mu), \Omega - B_{\mu'})$$

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